HAOMING CHEN

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EDUCATION

University of California, Berkeley Expected May 2028
Intended Majors: Computer Science, Mathematics GPA: 4.0/4.0
Relevant Coursework: Data Structures and Algorithms, Principles of Data Science, Deep Neural Networks, Deep Learning for Computer Vision, Large Language Model Agents

SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, HTML/CSS, MATLAB, Shell
Frameworks & Tools: PyTorch, TensorFlow, NumPy, Pandas, CUDA, DDP, Kubernetes, Git, Linux, Docker, Firebase, React, Next.js, Supabase, Agentic frameworks

EXPERIENCE

Software Engineering Intern, Databricks San Francisco, CA
AI Custom Model Training Team May 2026 – Present

Worked on the serverless GPU platform and AI runtime, contributing to the release of the companies' first CLI tool used by customers for model deployment and infrastructure management.

Implemented an end-to-end profiling and optimization framework within Databricks Workspace, enabling automated PyTorch profiling, trace visualization, and performance optimization through a unified interface.

Machine Learning Researcher, Berkeley Artificial Intelligence Research (BAIR) Berkeley, CA
Advised by Prof. Trevor Darrell and Dr. David Chan Jun 2025 – Present

Worked on the visual latent reasoning framework on text only domains, submitted paper at EMNLP 2025.

Developed a unified, reproducible implementation of the ViLex Visual Lexicon, integrating it with the Bagel vision language model to improve alignment between image reconstruction quality and visual understanding.

Machine Learning Researcher, Lawrence Berkeley National Laboratory Berkeley, CA
NERSC Computing Group Jan 2025 – Aug 2025

Designed scalable PyTorch data-loading pipelines supporting more than ten heterogeneous particle-physics data formats and over 100 million jet events, enabling large-scale foundation-model training. Developed a Mixture-of-Experts conditional tokenizer for jet-level representation learning, achieving state-of-the-art performance.

Software Engineer Intern, WeLeap AI San Francisco, CA
SkyDeck-backed Series A startup Jun 2025 – Sep 2025

Built a production-scale multi-agent financial assistant using the Gemini SDK and Firebase, capable of processing multimodal user inputs. Optimized the CQRS-based inference pipeline and streaming architecture, reducing response latency by 40% and improving end-to-end interaction speed by over 80%.

PROJECTS

Interpretability Research in Language Models on nationality bias — Inspired by the assistant axis research done at Anthropics, conducted experiments using various methods to analyze the formation of 'nationality axis' in language model and their alignments.

Latent Preference Formation in Language Models — Conducted experiments analyzing how data and post-training methods induces latent preference shifts in language models, employing behavioral benchmarks, probing techniques, and sparse autoencoders (SAEs) as tools for systematic safety and alignment evaluation.

Werewolf Agent Benchmark — Developed a multi-agent evaluation framework for measuring social intelligence in large language models using the Werewolf (Mafia) game. Implemented green-agent orchestration, role-aware white agents, vector stores, and benchmark hosts within an A2A architecture.